

Software Test Plan (STP) - Electronics Store Inventory Management System

Project: Electronics Store Inventory Management System

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Electronics Store Inventory Management System (EIMS) v1.0. It outlines the testing objectives, scope, test approach, environment, schedule, deliverables, responsibilities, risks, assumptions, suspension/resumption criteria, test case traceability, metrics, and approvals for the system.

Scope: Testing covers the EIMS web application features specified in the SRS: component inventory management, adding components, deleting/discontinuing components, availability reporting, dashboard functions, authentication and role-based access control, audit logging of inventory changes, and the measurable non-functional and security requirements. The system is intended for use by Admin/Store Manager and Sales/Store Staff roles.

The SRS describes EIMS as a standalone web-based application consisting of a front-end interface, a back-end REST API server, and a relational/NoSQL database. Testing therefore considers UI behaviour, API behaviour, persistence, role permissions, report correctness, dashboard consistency, security controls, usability, performance, scalability, maintainability, and desktop/tablet responsiveness.

Features outside the stated product scope are not treated as implemented EIMS requirements. In particular, the SRS excludes point-of-sale billing integration with external payment gateways and supplier-side procurement systems, although future interfaces may be considered.

References: Electronics Store Inventory Management System SRS v1.0 dated 06-09-2026; the SRS requirements and acceptance criteria in Sections 4, 5, 5.1, 6 and 8.

Definitions: EIMS (Electronics Store Inventory Management System), SRS (Software Requirements Specification), UI (User Interface), API (Application Programming Interface), DB (Database), SKU (Stock Keeping Unit), CRUD (Create/Read/Update/Delete), RBAC (Role-Based Access Control), JWT (JSON Web Token), ORM/ODM, REST (Representational State Transfer), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix).

- The SRS identifies functional requirement IDs EIMS-F-001 through EIMS-F-022.
- The SRS identifies non-functional requirement IDs EIMS-NF-001 through EIMS-NF-006.
- The SRS identifies security requirement IDs EIMS-SR-001 through EIMS-SR-006.

2. Test Items

- Components Inventory Management module – inventory list, search, filter, sort, and authorized update.
- Add Component module – mandatory fields, SKU generation, validation, duplicate prevention, and optional image upload.
- Delete Component module – authorized deletion, confirmation, role restriction, and audit logging.
- Availability Report module – classification, filtering, export, and low-stock highlighting.
- Dashboard module – summary widgets, charts, role-based views, and real-time low-stock alerts.
- Authentication and authorization functions – username/password authentication, RBAC, session expiry, and protected API access.
- Inventory audit logging – verification of recorded user, timestamp, and component information for deletion events.
- REST API and database interaction supporting component, category, supplier, and user data.
- Responsive web UI on desktop and tablet viewports.
- Security controls including password hashing, HTTPS/TLS 1.2+, input validation/sanitization, injection protection, and XSS protection.

3. Features to be Tested

Features mapped directly to the EIMS SRS requirement IDs are listed below. The test plan expands the SRS acceptance criteria into testable checks without introducing additional product requirements.

Req ID	Requirement to be verified	Type	Priority	Source/Stakeholder	Test ref
EIMS-F-001	Display active inventory list with name, SKU, category, quantity, unit price, supplier and reorder level.	Functional	High	Store Manager	TC-Inv-01
EIMS-F-002	Search components by name, SKU or category; matching results within 2 seconds.	Functional	High	Store Staff	TC-Inv-02
EIMS-F-003	Filter components by category, supplier and stock status.	Functional	Medium	Store Manager	TC-Inv-03

EIMS-F-004	Sort component list by quantity, price or date added.	Functional	Medium	Store Staff	TC-Inv-04
EIMS-F-005	Authorized users update quantity, price, supplier and reorder level; changes persist immediately.	Functional	High	Store Manager	TC-Inv-05
EIMS-F-006	Add component using mandatory name, category, quantity, unit price and supplier fields.	Functional	High	Store Manager	TC-Add-01
EIMS-F-007	Auto-generate unique, non-reused component ID/SKU.	Functional	High	System	TC-Add-02
EIMS-F-008	Validate mandatory fields and reject missing/invalid data with clear messages.	Functional	High	QA	TC-Add-03
EIMS-F-009	Reject duplicate component entries with the same SKU.	Functional	Medium	Store Manager	TC-Add-04
EIMS-F-010	Allow optional component image upload; image stored and displayed; maximum 5MB.	Functional	Low	Store Staff	TC-Add-05
EIMS-F-011	Allow authorized users to delete a component from inventory.	Functional	High	Store Manager	TC-Del-01
EIMS-F-012	Show confirmation prompt before permanent deletion.	Functional	High	QA	TC-Del-02
EIMS-F-013	Restrict delete access to Admin/Store Manager roles.	Functional	High	Security	TC-Del-03
EIMS-F-014	Record audit log entry containing user, timestamp and component for every deletion.	Functional	Medium	Compliance	TC-Del-04
EIMS-F-015	Generate real-time availability report classifying in-stock, low-stock and out-of-stock.	Functional	High	Store Manager	TC-Rep-01
EIMS-F-016	Filter availability report by category, supplier or date range.	Functional	Medium	Store Staff	TC-Rep-02
EIMS-F-017	Export availability report as PDF, Excel or CSV; exported data matches on-screen view.	Functional	Medium	Store Manager	TC-Rep-03
EIMS-F-018	Visually highlight components at or below reorder threshold.	Functional	Medium	Store Manager	TC-Rep-04
EIMS-F-019	Dashboard shows total components, total stock value, low-stock count and recently added items.	Functional	High	Store Manager	TC-Dash-01
EIMS-F-020	Dashboard displays stock-level/category and stock-trend charts; charts update with inventory changes.	Functional	Medium	Store Manager	TC-Dash-02
EIMS-F-021	Provide role-based dashboard views: full Admin view and read-only Staff summary.	Functional	Medium	Security	TC-Dash-03
EIMS-F-022	Display real-time low-stock alerts/notifications; alert appears within 5 seconds of threshold crossing.	Functional	Medium	Store Manager	TC-Dash-04

3.1 Non-functional requirements

Req ID	Requirement to be verified	Category	Priority	Acceptance / Test ref
EIMS-NF-001	Dashboard and inventory list pages load within 3 seconds for 90% of requests under normal load.	Performance	High	TC-Perf-01
EIMS-NF-002	System provides 99.5% monthly availability excluding scheduled maintenance.	Reliability	High	Ops reports

EIMS-NF-003	New staff member can add or search a component within 5 minutes of first use without formal training.	Usability	Medium	TC-UX-01
EIMS-NF-004	Support at least 10,000 components and 20 concurrent users without performance degradation.	Scalability	Medium	TC-Scale-01
EIMS-NF-005	Codebase is modular with separate UI, API and data-access layers and is documented for maintenance.	Maintainability	Medium	Code review checklist
EIMS-NF-006	System is responsive and usable on desktop and tablet screen sizes.	Portability/Usability	Low	TC-UX-02

3.2 Security requirements

Req ID	Requirement to be verified	Type	Priority	Acceptance / Test ref
EIMS-SR-001	Authentication required before access to any inventory function; unauthenticated protected requests rejected (401).	Security	High	TC-Sec-01
EIMS-SR-002	User passwords stored as salted one-way hashes, never plaintext.	Security	High	TC-Sec-02
EIMS-SR-003	RBAC distinguishes Admin/Manager and Staff; Staff access to delete/admin endpoints returns 403.	Security	High	TC-Sec-03
EIMS-SR-004	TLS 1.2+ enforced for all client-server communication; plain HTTP redirected/rejected.	Security	High	TC-Sec-04
EIMS-SR-005	Inactive user session automatically terminates after 15 minutes and requires re-login.	Security	Medium	TC-Sec-05
EIMS-SR-006	All user input validated and sanitized to prevent SQL/NoSQL injection and XSS; automated scan has no critical findings.	Security	High	TC-Sec-06

4. Features Not to be Tested

- Point-of-sale (billing) integration with external payment gateways, because the SRS explicitly excludes this from the current scope.
- Supplier-side procurement systems, because the SRS excludes supplier-side procurement from the current scope.
- Optional/future supplier low-stock notification functionality as a current implemented requirement; the SRS describes the Supplier role and notification as optional/future.
- Barcode scanner and printer behaviour as mandatory product features; the SRS identifies these hardware interfaces as optional.
- Specific technology implementation details not selected by the development team, such as whether Django/DRF or the MERN stack is used. Testing will verify the required externally observable behaviour and REST API architecture rather than imposing a technology choice.
- Third-party export-service internals or external notification-service internals; only the EIMS-facing behaviour specified in the SRS is considered when those optional interfaces are implemented.

5. Test Approach / Strategy

The strategy follows the structure and acceptance criteria in the EIMS SRS. Testing will be performed at module, integration, system and acceptance levels so that each requirement is verified both independently and in the workflows where it is used.

Levels

- Unit tests (module-level): Validate inventory operations, validation rules, SKU generation, threshold classification, calculation logic for dashboard values, and security-related utility functions.
- Integration tests (module/API/database): Verify communication between the web UI, REST API and database, including persistence of component updates, report data, authentication tokens and audit-log entries.
- System tests (end-to-end): Execute complete Admin/Store Manager and Store Staff workflows through the web application, including login, inventory viewing, adding, updating, deleting, reporting and dashboard use.
- Acceptance tests (UAT): Execute the acceptance suites stated in the SRS: Inventory Management, Add Component, Delete Component, Availability Report, Dashboard, Security and Usability.

Types

- Functional testing – verify every EIMS-F-001 through EIMS-F-022 requirement against its SRS acceptance criteria.
- Regression testing – repeat affected functional and security tests after fixes to inventory, authentication, report and dashboard modules.
- Performance testing – measure inventory/search/report/dashboard response behaviour against the SRS time limits.
- Scalability/load testing – use at least 10,000 components and 20 concurrent sessions for EIMS-NF-004.
- Usability testing – verify the five-minute task completion criterion for a new staff member and responsive use on desktop/tablet.
- Security testing – authentication, authorization, password storage, HTTPS/TLS, session expiry, input sanitization, injection and XSS checks.
- Data validation testing – verify inventory, report and dashboard values against known test datasets and threshold values.

Entry Criteria

- A stable EIMS build is delivered to the test environment.
- Required component, category, supplier and user test data are available.
- Admin/Manager and Staff test accounts are available.
- The database and REST API are running and reachable from the test environment.
- Required test tools and browsers are installed.
- The test environment supports the required desktop and tablet viewport checks.

Exit Criteria

- 100% of planned test cases for the agreed test cycle are executed.
- All high-priority functional requirements are implemented and verified, consistent with the SRS acceptance exit criterion.
- No critical NFR failures remain.
- RTM shows all planned acceptance test cases passed.
- Blocking defects affecting core inventory, authentication, authorization, reporting or dashboard functionality are resolved or formally accepted by the project team/guide.

5.1 Security Validation

Security validation is based directly on the security objectives and six security requirements in the SRS. The focus is on protecting inventory and user-account data and ensuring that only authenticated and authorized personnel can perform protected inventory actions.

- EIMS-SR-001 – Authentication: attempt to access protected inventory pages and endpoints without a valid authenticated session. Verify rejection with HTTP 401 for protected requests.
- EIMS-SR-001 – Valid credentials: verify that a valid username/password permits access to the inventory functions allowed for the account.
- EIMS-SR-002 – Password storage: inspect the user record in the database/test persistence layer and verify that the password is represented by a salted one-way hash rather than plaintext.
- EIMS-SR-003 – RBAC: use both Admin/Manager and Staff accounts. Verify Staff cannot invoke delete/admin endpoints and receives HTTP 403 where specified.
- EIMS-SR-003 – UI permission consistency: verify that Staff dashboard/inventory views do not expose controls that are restricted to Admin/Manager.
- EIMS-SR-004 – HTTPS: verify client-server communication uses HTTPS and TLS 1.2 or later. Plain HTTP requests must be redirected or rejected as specified.
- EIMS-SR-005 – Session expiry: leave an authenticated session inactive for 15 minutes and verify that the token/session expires and a new login is required.

- EIMS-SR-006 – Input validation/sanitization: submit malicious or malformed values in component and user input fields and verify that SQL/NoSQL injection and XSS payloads are not executed or accepted.
- EIMS-SR-006 – Automated scan: run a security scan such as OWASP ZAP against the test deployment and verify that no critical injection/XSS findings are reported.
- Delete authorization: combine EIMS-F-013 with EIMS-SR-003 by attempting deletion using a Staff account and confirming access is denied.
- Audit logging: after an authorized deletion, verify that the recorded user, timestamp and component correspond to the actual event, as required by EIMS-F-014.

6. Test Environment

Hardware

- Desktop/laptop system capable of running a modern browser and the EIMS client application.
- Tablet or tablet-sized viewport/device for responsive UI verification.
- Standard keyboard, mouse and touchscreen input where available.
- Barcode scanner may be used for optional lookup/entry testing if the implementation includes the optional interface.
- Printer may be used for optional report printing/export workflow if implemented.

Software

- EIMS web application v1.0 test build.
- Back-end REST API exposing JSON over HTTPS.
- Relational/NoSQL database selected by the development team.
- Modern browser: Chrome, Firefox or Edge, as stated in the SRS.
- Test data containing components, categories, suppliers and user accounts.
- For export testing, the implemented PDF/Excel/CSV generation capability.
- For security validation, a tool such as OWASP ZAP where applicable.
- For performance/load tests, a suitable load-testing tool capable of measuring request response times and concurrent sessions.

Test Data

- Inventory dataset containing component name, SKU, category, quantity, unit price, supplier and reorder level.
- Components representing in-stock, low-stock and out-of-stock conditions.
- Admin/Store Manager and Sales/Store Staff test accounts.
- Records for newly added components, updated components and recently added items.
- Data set of at least 10,000 components and 20 concurrent sessions for the scalability requirement.

Environment Verification

- Confirm the application loads through a supported browser before execution begins.
- Confirm the inventory list can retrieve active components from the database.
- Confirm category and supplier master data are available before tests depending on filters or component creation are executed.
- Confirm Admin/Manager and Staff credentials are valid and mapped to the correct roles.
- Confirm HTTPS is enabled before security and protected API tests are executed.
- Confirm database persistence is available so update, deletion, audit-log and dashboard consistency can be verified.
- Confirm the test environment can support the 20 concurrent-session scalability test without unrelated infrastructure limitations.
- Record browser, environment and build version information in execution logs so failures can be reproduced.

Test Data Preparation by Requirement Group

Test area	Data / setup	Requirement mapping
Inventory	Active components with complete attributes; different quantities, prices, categories and suppliers.	F-001 to F-005
Add Component	Valid component; missing fields; invalid fields; duplicate SKU attempt; image files below/at/above 5MB.	F-006 to F-010
Delete	Existing deletable component; Admin/Manager account; Staff account; deletion event for audit verification.	F-011 to F-014
Availability	Components above, at and below reorder level; different categories/suppliers; date-range records.	F-015 to F-018
Dashboard	Inventory data producing known total count, stock value, low-stock count and recently added items; category/trend data.	F-019 to F-022

Performance/Scale	Normal-load requests; 10,000-component dataset; 20 concurrent sessions.	NF-001, NF-004
Usability/Portability	New staff test users; desktop ≥1280px and tablet ≥768px viewports.	NF-003, NF-006
Security	Admin and Staff accounts; protected endpoints; inactive session; HTTPS; malicious input strings.	SR-001 to SR-006

7. Test Schedule

Milestone	Start	End	Activity
Test case design	18-Sep-2026	20-Sep-2026	Prepare functional, NFR and security test cases mapped to SRS IDs.
Test data preparation	19-Sep-2026	21-Sep-2026	Prepare inventory, category, supplier and role-based test data.
Environment setup	20-Sep-2026	22-Sep-2026	Deploy stable build, configure database/API/browser environment.
Functional execution	23-Sep-2026	28-Sep-2026	Execute Inventory, Add, Delete, Availability Report and Dashboard tests.
Security execution	26-Sep-2026	29-Sep-2026	Execute SR-001 to SR-006 security validation.
Performance/scale/usability	29-Sep-2026	01-Oct-2026	Execute NF-001 to NF-006 applicable measurements and reviews.
Regression testing	02-Oct-2026	04-Oct-2026	Re-run affected cases after defect fixes.
UAT / acceptance	05-Oct-2026	07-Oct-2026	Run SRS acceptance suites and verify exit criteria.
Test summary and sign-off	08-Oct-2026	09-Oct-2026	Prepare final execution summary, RTM status and approvals.

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated where applicable)
- Test Scripts for functional, security, performance and usability checks where automation is selected
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (RTM) updated with execution status
- Performance/load results for EIMS-NF-001 and EIMS-NF-004
- Security validation results for EIMS-SR-001 through EIMS-SR-006
- Usability and responsive UI results for EIMS-NF-003 and EIMS-NF-006
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Hemavathi E	Prepare test plan, coordinate test execution, monitor requirement coverage, review defects and coordinate final test reporting.
Test Engineer	Jayashree D	Design and execute test cases, prepare test data, record actual results, attach evidence and log defects.
Developer	Krishitha Kankanala	Support test environment setup, investigate defects, provide fixes and participate in defect triage/regression.
Product Owner / Project Team Lead	Karan Maheshwari	Review acceptance results, coordinate project-level decisions and support readiness/sign-off.
Project Guide / Evaluator	Prof. M. Sheela Devi	Review the project/test artefacts and provide academic/project guidance and acceptance feedback.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds and execute test design/data preparation in parallel so execution can begin when the stable build is available.
Test environment or database unavailable	Maintain a repeatable local/test deployment configuration and preserve prepared test data and test cases.
Incorrect or incomplete master data	Prepare category and supplier master data before dependent Inventory, Add Component, Filter and Report tests.
Role configuration error	Maintain separate Admin/Manager and Staff test accounts and verify role permissions before RBAC execution.

Optional feature not implemented	Mark optional interfaces such as barcode scanner, printer, notification service or export capability according to actual implementation; do not treat an unimplemented optional interface as a mandatory failure unless the team has included it in the release.
Large test dataset affects environment performance	Prepare the 10,000-component dataset in advance and record environment limitations separately from application results.
Security test payloads affect test data	Use isolated test accounts and non-production test data for injection/XSS and session-expiry checks.
Defects in shared inventory data affect later tests	Use controlled test records and reset/restore data between tests where required.

11. Assumptions & Dependencies

- The EIMS SRS v1.0 is the baseline for functional, non-functional and security verification.
- The application exposes the REST API architecture required by the SRS.
- A relational or NoSQL database is available for persistence of component, category, supplier, transaction and user information as applicable to the implementation.
- Category and supplier master data are populated before tests that depend on them.
- Admin/Manager and Staff test accounts are provided and correctly configured.
- The test environment has a modern supported browser such as Chrome, Firefox or Edge.
- A stable LAN or internet connection is available for the web application where required.
- HTTPS/TLS 1.2+ configuration is available for security verification.
- Optional interfaces are tested only if they are included in the implementation under test.
- PDF/Excel/CSV export is tested when the corresponding export capability is included in the release.
- Security scanning tools such as OWASP ZAP are available if used for EIMS-SR-006.
- Performance results are interpreted against the environment in which the production-like test is performed.
- The 99.5% monthly availability requirement depends on availability/uptime measurement from the operational environment.
- The academic project timeline may require schedule adjustment while maintaining the stated test activities and acceptance criteria.

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment or application is unavailable for an extended period that prevents meaningful execution.
- The build is too unstable to execute the planned core test cases, particularly when inventory, authentication or API access is blocked.
- Database corruption or missing master data prevents dependent tests from producing valid results.
- A critical security issue is discovered that requires the environment to be isolated before further security testing.
- A blocking defect prevents execution of a substantial portion of the current test suite.

Resume testing if:

- Blocking defects affecting the planned test sequence are resolved and the corrected build is available.
- The environment and database are stabilized and the required test data is restored.
- Authentication and role configuration are working sufficiently to continue protected functional testing.
- Security test execution can proceed in the isolated test environment.
- A smoke test confirms that the corrected build supports the previously blocked workflow.

13. Test Case Management & Traceability

The RTM ensures that every SRS requirement selected for verification is mapped to one or more test cases or review activities. The test case IDs below follow the naming used in the SRS, so execution results can be linked directly to the requirement and acceptance criterion.

Req ID	Requirement short	Section / Design Spec	Module	Test case(s)
EIMS-F-001	View inventory list	4.1 / DS-Inv-01	InventoryModule	TC-Inv-01
EIMS-F-002	Search by name/SKU/category	4.1 / DS-Inv-02	InventoryModule	TC-Inv-02
EIMS-F-003	Filter by category/supplier/status	4.1 / DS-Inv-03	InventoryModule	TC-Inv-03
EIMS-F-004	Sort by quantity/price/date added	4.1 / DS-Inv-04	InventoryModule	TC-Inv-04
EIMS-F-005	Update component details	4.1 / DS-Inv-05	InventoryModule	TC-Inv-05
EIMS-F-006	Add new component	4.2 / DS-Add-01	InventoryModule	TC-Add-01
EIMS-F-007	Unique SKU generation	4.2 / DS-Add-02	InventoryModule	TC-Add-02
EIMS-F-008	Mandatory-field validation	4.2 / DS-Add-03	InventoryModule	TC-Add-03
EIMS-F-009	Duplicate SKU prevention	4.2 / DS-Add-04	InventoryModule	TC-Add-04
EIMS-F-010	Optional image upload	4.2 / DS-Add-05	InventoryModule	TC-Add-05
EIMS-F-011	Delete component	4.3 / DS-Del-01	InventoryModule	TC-Del-01
EIMS-F-012	Deletion confirmation	4.3 / DS-Del-02	InventoryModule	TC-Del-02
EIMS-F-013	Delete role restriction	4.3 / DS-Del-03	AuthModule	TC-Del-03

EIMS-F-014	Deletion audit log	4.3 / DS-Del-04	AuditModule	TC-Del-04
EIMS-F-015	Generate availability report	4.4 / DS-Rep-01	ReportModule	TC-Rep-01
EIMS-F-016	Filter availability report	4.4 / DS-Rep-02	ReportModule	TC-Rep-02
EIMS-F-017	Export report	4.4 / DS-Rep-03	ReportModule	TC-Rep-03
EIMS-F-018	Highlight low-stock components	4.4 / DS-Rep-04	ReportModule	TC-Rep-04
EIMS-F-019	Dashboard summary widgets	4.5 / DS-Dash-01	DashboardModule	TC-Dash-01
EIMS-F-020	Dashboard charts	4.5 / DS-Dash-02	DashboardModule	TC-Dash-02
EIMS-F-021	Role-based dashboard	4.5 / DS-Dash-03	DashboardModule/AuthModule	TC-Dash-03
EIMS-F-022	Real-time low-stock alert	4.5 / DS-Dash-04	DashboardModule	TC-Dash-04

Req ID	Requirement short	Section / Design Spec	Module	Test case(s)
EIMS-NF-001	Page load performance	5 / DS-Perf-01	WebUI / API	TC-Perf-01
EIMS-NF-002	Monthly availability	5 / DS-Rel-01	Operations	Ops reports
EIMS-NF-003	New staff usability	5 / DS-UX-01	WebUI	TC-UX-01
EIMS-NF-004	10,000 components / 20 users	5 / DS-Scale-01	WebUI / API / DB	TC-Scale-01
EIMS-NF-005	Modular codebase/documentation	5 / DS-Maint-01	Codebase	Code review checklist
EIMS-NF-006	Desktop/tablet responsiveness	5 / DS-UX-02	WebUI	TC-UX-02
EIMS-SR-001	Authentication required	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-01
EIMS-SR-002	Salted one-way password hash	5.1.2 / DS-Sec-02	AuthModule/DB	TC-Sec-02
EIMS-SR-003	Role-based access control	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03
EIMS-SR-004	TLS 1.2+ HTTPS	5.1.2 / DS-Sec-04	Web/API	TC-Sec-04
EIMS-SR-005	15-minute inactive session termination	5.1.2 / DS-Sec-05	AuthModule	TC-Sec-05
EIMS-SR-006	Input sanitization / injection / XSS	5.1.2 / DS-Sec-06	Web/API	TC-Sec-06

RTM status convention: N = Not executed, P = Passed, A = Accepted/closed as applicable to the project's test reporting. Initial status for the draft plan is N until the corresponding test case or review is executed.

Requirement-to-test grouping

- TC-Inv-01 through TC-Inv-05 cover EIMS-F-001 through EIMS-F-005.
- TC-Add-01 through TC-Add-05 cover EIMS-F-006 through EIMS-F-010.
- TC-Del-01 through TC-Del-04 cover EIMS-F-011 through EIMS-F-014.
- TC-Rep-01 through TC-Rep-04 cover EIMS-F-015 through EIMS-F-018.
- TC-Dash-01 through TC-Dash-04 cover EIMS-F-019 through EIMS-F-022.
- TC-Perf-01, Ops reports, TC-UX-01, TC-Scale-01, Code review checklist and TC-UX-02 cover EIMS-NF-001 through EIMS-NF-006.
- TC-Sec-01 through TC-Sec-06 cover EIMS-SR-001 through EIMS-SR-006.

14. Test Metrics & Reporting

Metrics collected

- % test cases executed – planned versus executed cases for the current cycle.
- % passed/failed – execution outcome of functional, NFR and security test cases.
- Defect density – defects recorded relative to the size of the tested module/test scope.
- Defect aging – time defects remain open from logging until closure.
- Requirement coverage – percentage of SRS requirements mapped to executed test cases/reviews.
- High-priority requirement coverage – status of High-priority functional and security requirements.
- Performance measurements – percentage of dashboard/inventory requests meeting the 3-second criterion and search/alert measurements where specified.
- Scalability result – whether the 10,000-component/20-concurrent-user test passes without performance degradation.
- Usability result – task completion time for the new staff member against the five-minute criterion.
- Security result – pass/fail status of authentication, hashing, RBAC, TLS, session expiry and injection/XSS checks.

Reports

- Daily execution status – cases executed, pass/fail count, blockers and newly logged defects.
- Defect reports – requirement/test case reference, steps to reproduce, expected result, actual result, severity/priority as agreed by the team, evidence and current status.
- Performance/load report – measured timings, request counts/concurrency and comparison against the SRS thresholds.
- Security validation report – test results for EIMS-SR-001 through EIMS-SR-006 and security scan findings.
- Final Test Summary Report – overall execution status, requirement coverage, open defects, NFR/security results, acceptance status and sign-off readiness.

Functional execution reporting

Test area	Test references	SRS mapping	Reported result
Inventory	TC-Inv-01 to TC-Inv-05	F-001 to F-005	Executed / Passed / Failed / Blocked
Add Component	TC-Add-01 to TC-Add-05	F-006 to F-010	Executed / Passed / Failed / Blocked
Delete Component	TC-Del-01 to TC-Del-04	F-011 to F-014	Executed / Passed / Failed / Blocked
Availability Report	TC-Rep-01 to TC-Rep-04	F-015 to F-018	Executed / Passed / Failed / Blocked
Dashboard	TC-Dash-01 to TC-Dash-04	F-019 to F-022	Executed / Passed / Failed / Blocked
Non-functional	TC-Perf-01, Ops, TC-UX-01, TC-Scale-01, review, TC-UX-02	NF-001 to NF-006	Measurement / Review result
Security	TC-Sec-01 to TC-Sec-06	SR-001 to SR-006	Passed / Failed / Findings

Acceptance readiness

- All High-priority functional requirements are implemented and verified.
- No critical NFR failures remain.
- The RTM shows the acceptance test cases as passed.
- The acceptance suites cover Inventory Management, Add Component, Delete Component, Availability Report, Dashboard, Security and Usability, as specified in the SRS.
- Final test reporting records any remaining non-blocking observations separately from failed requirements.

15. Approvals

Role	Name	Signature / Date
QA Lead	Hemavathi E	
Dev Lead	Krishitha Kankanala	
Product Owner / Project Team Lead	Karan Maheshwari	
Project Guide / Evaluator	Prof. M. Sheela Devi	